

Mitigating Class-Imbalance Damage on BRACS: CUDA and Logit Adjustment Recover up to Half, No Method Closes the Gap

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Abstract

Class prevalence alone costs 7.65 points of balanced accuracy on BRACS at an imbalance ratio of 100 under a frozen Virchow2 encoder and a logistic-regression readout. This report asks how much of that damage six established mitigation methods give back once the encoder adapts through low-rank updates (LoRA). On the 30 split-draw shards of the prevalence study, the unmitigated cross-entropy (CE) model is trained on a balanced allocation and, together with every method over a grid of strengths, on allocations with ratios 50 and 100 and with BRACS’s native class shares; each method’s strength is selected per shard on validation patients only. Under LoRA, CE loses 4.95 points at ratio 100 and 3.54 points at ratio 50, while at native shares its 0.85-point loss cannot be distinguished from zero. At ratio 100, six of eight method variants meet the pre-stated recovery criterion of at least one point with an interval above zero. A curriculum of data augmentation (CUDA) recovers the most, 2.59 points or about half of the damage, followed by train-time and post-hoc logit adjustment at 1.96 and 1.94 points, or about two-fifths; the two logit-adjustment forms are the only methods that lift tail-class recall substantially. No paired share interval reaches full recovery. CUDA is also the only method that gains at native shares, by 1.21 points against a damage that cannot be distinguished from zero, so part of its recovery likely reflects augmentation that helps regardless of imbalance. Apart from CUDA, methods that re-fit only the classifier on a frozen encoder do as well as methods trained end to end. The smaller LoRA damage reflects a balanced model about three points less accurate than the frozen readout, not a more robust imbalanced one.

1 Motivation

The prevalence study of this series isolates class prevalence from patient shortage at fixed patient support. On BRACS it finds 7.65 points of balanced-accuracy damage at an imbalance ratio of $\rho = 100$, against 2.91 points on TCGA-UT under the same design (Queisler, 2026d,e). A follow-up split the BRACS damage into a prior channel and a support channel of similar size and found that post-hoc logit adjustment on the frozen readout recovers only about a quarter of it (Queisler, 2026a). Both studies keep the encoder frozen, so neither tests the established long-tailed methods that also shape the representation.

A companion report defines six such methods for patch classification under a Virchow2 encoder adapted only through LoRA: balanced sampling and a curriculum of data augmentation (CUDA) at the data level, logit adjustment and DisAlign at the algorithm level, and classifier re-training (cRT) and Gaussian clouded logit (GCL) adjustment as hybrids (Queisler, 2026b). For the other damage source this series studies, patient shortage, six earlier mitigation studies found that none of their corrections worked on BRACS (Queisler, 2026c). This report asks: *once the encoder can adapt, how much of BRACS’s prevalence damage does each method recover, and does recovery differ between methods trained end to end and methods that freeze the encoder and re-fit only the classifier?*

2 Design

The shard design is that of the prevalence study (Queisler, 2026d): three fixed patient-disjoint BRACS splits, each drawn ten times, with $G = 10$ patients per lesion class sampled from those with at least 160 training patches and a total budget of $T = 7 \cdot G \cdot 32 = 2,240$ patches held fixed across all allocations of a draw. Every allocation reuses the prevalence study’s patient draws and class-rank permutation, so the 30 split-draw shards differ from that study only in the classifier. Training reads decoded patch images, because every method adapts the encoder: a rank-8 LoRA update on the attention projections plus a linear head (a cosine head for GCL), as defined by Queisler (2026b).

2.1 Allocations and methods

Each shard’s patch budget is divided among the seven classes in four ways, all taken from the prevalence study’s exponential allocation profile. The *balanced* allocation ($\rho = 1$) trains only CE, which serves as the *undamaged reference*. Three imbalanced allocations, with ratios $\rho = 50$ and $\rho = 100$ and with BRACS’s native class shares (N, realized ratio about 6.7), each train CE as their *damaged reference* and every mitigation method over the grid of Table 1. The two ratios show whether recovery grows with severity; the native shares ask whether mitigation matters at the imbalance BRACS actually has.

Ratio 100 is close to the most severe imbalance this design can express. At $\rho = 100$ the rarest class receives about 12 of the 2,240 patches, roughly one per patient. Beyond $\rho \approx 125$ it would receive fewer patches than its ten patients, so the tail class could no longer draw on all its patients and the design would mix prevalence with patient shortage.

Methods fall into two training families. *End-to-end* methods (balanced sampling, CUDA, train-time logit adjustment, GCL) train the LoRA update and head together. CUDA (Ahn et al., 2023) augments image patches before the encoder, with a per-class strength that rises as the current model learns the class, so it can only act end to end. *Classifier-only* methods freeze an end-to-end model’s encoder and re-fit only classifier parameters: post-hoc logit adjustment, DisAlign, and cRT start from the allocation’s CE model, and GCL’s classifier-only form, its second training stage, starts from the end-to-end GCL model of the same σ . Logit adjustment and GCL therefore each appear once per family, giving eight method variants.

2.2 Training length and fit count

End-to-end training length was fixed before the grid by a pilot: CE on the balanced and the $\rho = 100$ allocation, one draw per split, trained for 150, 300, 600, and 1,200 steps. The rule selects the smallest step count within 0.5 points of the best mean validation balanced accuracy at $\rho = 100$. The four candidates reached 31.87, 34.22, 34.93, and 33.42 percent, so 600 steps (batch 256, learning rate 10^{-4} , cosine schedule) were used for all end-to-end fits; classifier-only fits train 200 steps. One length serves all allocations, so that a recovery difference between allocations cannot come from training length. It was chosen at the most severe allocation and not re-tuned at $\rho = 50$ or native shares. Per shard, the grid holds 35 end-to-end and 38 classifier-only configurations, for 1,050 and 1,140 fits over the 30 shards. Every configured fit completed and entered the analysis.

3 Evaluation

The primary endpoint is test patient-macro balanced accuracy (BA), the mean of per-class patch recalls in which every test patient of a class carries equal weight (Queisler, 2026d). Estimates pool the 30 shards with equal weight per split. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates.

Training	Method	Strength	Grid
End to end	CE (reference)	–	–
End to end	Balanced sampling	s	0.5, 1.0
End to end	CUDA	p_{aug}	0.25, 0.5, 1.0
End to end	Logit adjustment (train-time)	τ	0.5, 1.0
End to end	GCL	σ	0.02, 0.05, 0.2 (N: also 0.5)
Classifier only	Logit adjustment (post hoc)	τ	0.5, 1.0, 1.5, 2.0
Classifier only	DisAlign	γ	0.5, 1.0, 1.5, 2.0 (N: also 3.0)
Classifier only	cRT	–	–
Classifier only	GCL	σ of its source	as GCL

Table 1: Strength grids of the imbalanced allocations. Balanced sampling spans the upper half of its defined range $s \in [0, 1]$, CUDA’s augmentation probability p_{aug} ends at its maximum of 1, and train-time τ stops at the full prior correction $\tau = 1$. CUDA’s acceptance threshold (0.6), check size (10 patches per strength), and maximum strength (10 operations) are fixed, and its per-class levels are updated every nine steps, one pass over the 2,240 training patches. The other grids bracket the source defaults and extend past any grid edge on which the validation selection would otherwise concentrate; at native shares the GCL and DisAlign grids reach further for that reason. γ is DisAlign’s reweighting exponent, written ρ in the methods report. The balanced allocation trains only CE.

All allocations and methods share the replicate weights, so every contrast is paired; intervals are 95 percent percentile intervals.

For each method, shard, and imbalanced allocation, the grid value with the best *validation* BA is selected; the test partition is never read for selection. With m^* the selected fit and a the allocation,

$$D_a = \text{BA}(\text{CE}, 1) - \text{BA}(\text{CE}, a), \quad R_m = \text{BA}(m^*, a) - \text{BA}(\text{CE}, a), \quad \text{share}_m = R_m / D_a, \quad (1)$$

where the share is computed per replicate. A method *recovers* a material part of the damage if $R_m \geq 1$ point and its interval lies above zero. It shows *full recovery* if the interval of $\text{BA}(m^*, a)$ contains the point estimate of $\text{BA}(\text{CE}, 1)$. Both thresholds were fixed before the fits.

Secondary readings are descriptive: head, body, and tail recall by assigned class rank; macro negative log-likelihood (NLL) and expected calibration error (ECE), raw and after validation temperature scaling, following the prevalence study’s recipe; the frequency of each selected strength; and the mean share of the end-to-end and classifier-only families. The prevalence study’s frozen-readout damage uses the same test patients but a different classifier, so it serves as an external reference without a paired interval.

4 Results

4.1 Damage under LoRA

The balanced CE model reaches 36.53 percent BA [33.82, 39.45]. It loses $D = 3.54$ points [1.93, 5.21] at $\rho = 50$ and 4.95 points [3.25, 6.74] at $\rho = 100$. At native shares the damage is 0.85 points [−0.31, 2.08], an interval that includes zero. All three damages are smaller than under the frozen readout, which lost 6.63, 7.65, and 4.05 points on the same test patients (Queisler, 2026d). The reason is the undamaged reference, not the imbalanced allocations: LoRA’s balanced model is about three points less accurate than the frozen readout (36.53 against 39.57 percent), while the imbalanced models land close to the frozen values (32.99 against 32.94 at $\rho = 50$, 31.57 against 31.92 at $\rho = 100$, 35.67 against 35.51 at N).

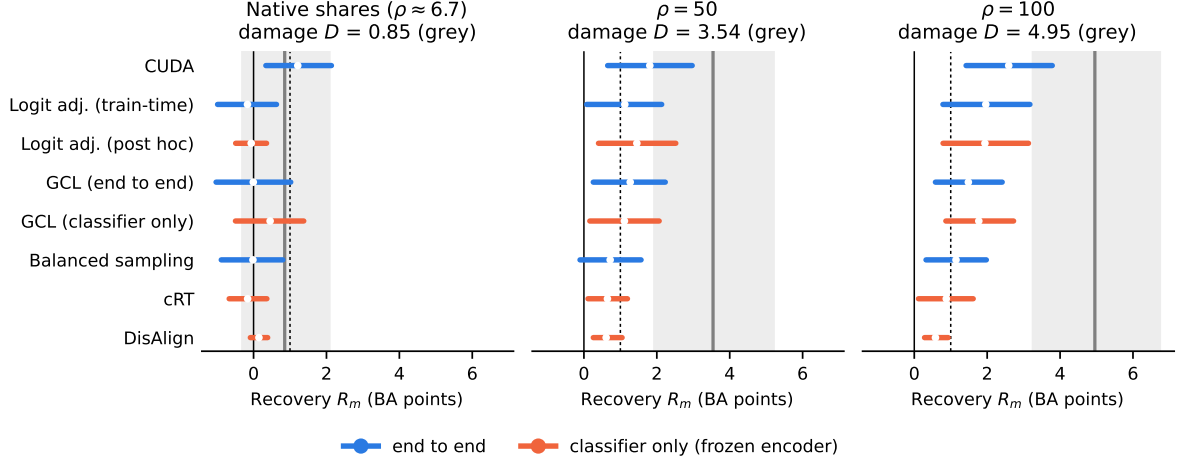


Figure 1: CUDA, logit adjustment, and GCL recover one to three points at $\rho = 50$ and $\rho = 100$, short of the damage; at native shares only CUDA moves. Paired recovery R_m with 95% intervals. The grey line and band mark the CE damage D with its interval, the solid line zero, and the dotted line the one-point threshold.

4.2 Recovery per method

Figure 1 and Table 2 give each method’s paired recovery. At $\rho = 100$, six of eight variants meet the recovery criterion: CUDA (2.59 points [1.42, 3.79]), train-time and post-hoc logit adjustment (1.96 and 1.94), classifier-only and end-to-end GCL (1.77 and 1.48), and balanced sampling (1.14). cRT (0.88) and DisAlign (0.58) recover a positive amount below the one-point threshold. CUDA’s share of the damage, 0.52 [0.29, 0.82], is the largest, followed by logit adjustment’s 0.40 [0.19, 0.60]; no share interval reaches 1, so part of the damage remains under every method. CUDA alone meets the full-recovery criterion at $\rho = 100$: the interval of its accuracy, [31.70, 36.81], just contains the balanced 36.53 percent, while its paired share places the remaining gap between about one-fifth and seven-tenths of the damage.

At $\rho = 50$ the ordering is the same with smaller gains. CUDA (1.81 points, share 0.51 [0.20, 0.91]), logit adjustment (both forms), and GCL (both forms) meet the recovery criterion, the latter four with shares of 0.31 to 0.41, and balanced sampling falls just short (0.72, interval includes zero). These five variants also meet the full-recovery criterion, because the interval of an unpaired accuracy is about five points wide, wider than the 3.54-point damage; their paired shares, whose intervals end below 0.92, show that the gap is not closed. At native shares, seven of eight recoveries lie between -0.17 and 0.45 points with an interval that includes zero. CUDA is the exception: it gains 1.21 points [0.34, 2.14] and reaches 36.89 percent, above the balanced CE model’s 36.53, although the damage it would recover, 0.85 points, cannot be distinguished from zero.

4.3 End-to-end versus classifier-only training

The two training families differ mainly through CUDA, which has no classifier-only form. The mean share is 0.36 for end-to-end and 0.26 for classifier-only methods at $\rho = 100$, and 0.35 and 0.27 at $\rho = 50$; without CUDA, the end-to-end mean at $\rho = 100$ is 0.31. The matched pairs agree most closely: post-hoc logit adjustment, applied to the CE model at inference without retraining, recovers as much as its train-time form (1.94 against 1.96 points at $\rho = 100$), and re-fitting GCL’s classifier adds 0.29 points to end-to-end GCL at $\rho = 100$ and loses 0.16 at $\rho = 50$. These family and pair comparisons are point estimates without intervals.

Method	$\rho = 50$ ($D = 3.54$)		$\rho = 100$ ($D = 4.95$)		N ($D = 0.85$)
	R_m	share	R_m	share	R_m
CUDA	1.81 [0.65, 2.97]	0.51 [0.20, 0.91]	2.59 [1.42, 3.79]	0.52 [0.29, 0.82]	1.21 [0.34, 2.14]
Logit adj. (train-time)	1.12 [0.08, 2.13]	0.32 [0.04, 0.58]	1.96 [0.79, 3.17]	0.40 [0.19, 0.60]	-0.17 [-0.99, 0.63]
Logit adj. (post hoc) [†]	1.45 [0.40, 2.52]	0.41 [0.15, 0.71]	1.94 [0.79, 3.13]	0.39 [0.18, 0.59]	-0.07 [-0.49, 0.36]
GCL	1.27 [0.26, 2.23]	0.36 [0.09, 0.66]	1.48 [0.58, 2.41]	0.30 [0.12, 0.51]	-0.01 [-1.03, 1.02]
GCL [†]	1.11 [0.16, 2.07]	0.31 [0.06, 0.60]	1.77 [0.87, 2.73]	0.36 [0.19, 0.56]	0.45 [-0.50, 1.38]
Balanced sampling	0.72 [-0.11, 1.57]	0.20 [-0.05, 0.43]	1.14 [0.32, 1.98]	0.23 [0.07, 0.41]	-0.02 [-0.89, 0.80]
cRT [†]	0.65 [0.11, 1.20]	0.18 [0.04, 0.33]	0.88 [0.11, 1.62]	0.18 [0.03, 0.30]	-0.16 [-0.67, 0.37]
DisAlign [†]	0.61 [0.26, 1.05]	0.17 [0.08, 0.32]	0.58 [0.28, 0.93]	0.12 [0.06, 0.18]	0.15 [-0.09, 0.39]

Table 2: Six variants recover at least one point at $\rho = 100$; no share interval reaches the full damage. Recovery R_m in BA points and share of the CE damage, with paired 95% intervals; bold marks the pre-stated recovery criterion ($R_m \geq 1$, interval above zero). [†]Classifier only, frozen encoder; unmarked rows train end to end. Shares at native shares are omitted because the damage there cannot be distinguished from zero.

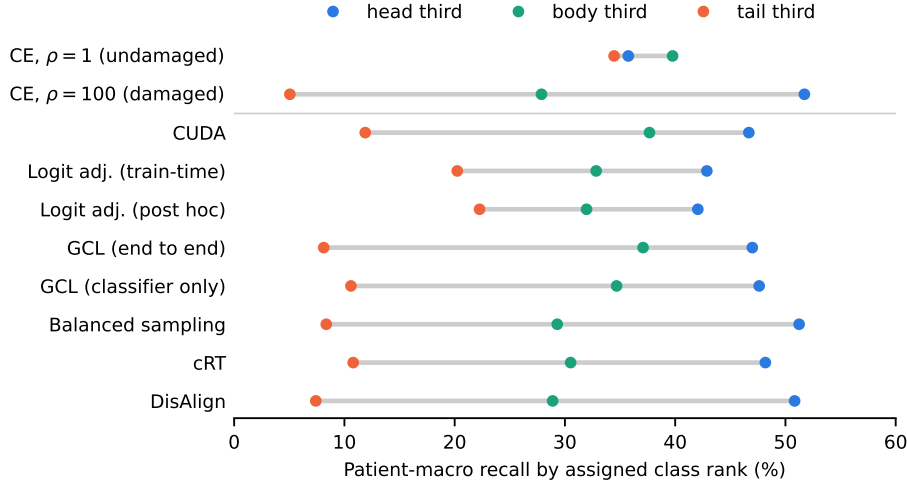


Figure 2: Only logit adjustment lifts tail recall substantially at $\rho = 100$; CUDA and GCL gain in the body, the other methods stay close to CE. Patient-macro recall of the head, body, and tail thirds of classes by assigned rank, averaged over shards.

4.4 Where recovery lands

At $\rho = 100$ the CE model keeps 5.1 percent recall in the tail third of classes by assigned rank, against 34.5 percent when balanced, and gains 16 points in the head (Figure 2). Only logit adjustment moves the tail substantially, to 20.2 (train-time) and 22.3 percent (post hoc), paid for by about nine points of head recall. CUDA and GCL recover through the body instead, 37.7 and 37.1 against 27.9 percent. CUDA raises the tail to 11.9 percent and gives up five points of head recall; GCL, balanced sampling, cRT, and DisAlign leave the tail at 7 to 11 percent. Rank-group recalls are point estimates.

4.5 Selected strengths

Validation selection mostly picks interior or theoretically motivated values. At $\rho = 100$, logit adjustment selects the full correction $\tau = 1$ in 23 of 30 shards in its train-time form, and $\tau = 1$ is also the post-hoc mode (15 of 30, with 0.5 in 11). Balanced sampling selects full balancing $s = 1$ in 22 of 30. GCL selects $\sigma = 0.05$ in 15 (end to end) and 16 (classifier only) of 30, and DisAlign selects $\gamma = 1$ or 1.5 in 22 of 30. CUDA’s selected augmentation probability rises with

the imbalance: $p_{\text{aug}} = 1$ wins 14 of 30 shards at $\rho = 100$, 13 of 30 at $\rho = 50$, and none at native shares, where 0.25 wins 17. The other selections at $\rho = 50$ resemble those at $\rho = 100$. At native shares the selections spread across the grid; DisAlign’s five values each win between 4 and 8 shards. With no resolved damage, validation accuracy is nearly flat in the strength, and the spread reflects noise rather than a preferred value.

4.6 Probability quality

Imbalance mainly worsens raw calibration, and temperature scaling removes most of it for every method. At $\rho = 100$ the CE model’s raw NLL rises from 2.89 to 3.49 nats and its raw ECE from 28.4 to 36.4 points; even the balanced LoRA model is strongly overconfident. CUDA has by far the lowest raw NLL, 2.25 nats [2.10, 2.41], and raw ECE, 20.8 points [16.7, 26.4], below even the balanced CE model (2.89 and 28.4); the two logit-adjustment forms follow (2.99 and 2.94 nats), and cRT (4.64) and balanced sampling (3.98) are highest. After scaling, every method lies between 1.71 (CUDA) and 1.82 nats, around CE’s 1.79 and close to the balanced 1.67, and scaled ECE falls from 12.3 for CE to between 8.8 and 12.3 points. The scaled intervals are unpaired and overlap: CUDA’s augmentation curbs raw overconfidence, but no method changes probability quality beyond what temperature scaling already does.

5 Discussion

How much does each method recover? At $\rho = 100$ six of eight method variants recover a material part of the damage, led by CUDA at about half and logit adjustment at about two-fifths. None closes the gap: no paired share interval reaches 1. The ranking is stable from $\rho = 50$ to $\rho = 100$, and the absolute gains grow with the damage.

Is CUDA’s gain an imbalance correction? Only in part, as far as this design can tell. CUDA is the one method that gains at native shares, where there is no resolved damage, and there its mitigated model exceeds the balanced CE model in point estimate. Its gain at native shares, 1.21 points, is about half of its gain at $\rho = 100$, so a part of the $\rho = 100$ recovery likely comes from augmentation that improves the LoRA model at any class distribution. The design fits CUDA only on imbalanced allocations and compares it with CE, so it cannot separate this general benefit from a correction of the imbalance. Two observations point to an imbalance-specific part as well: the selected augmentation probability rises with the ratio, and CUDA’s recovery grows from 1.21 at native shares to 2.59 points at $\rho = 100$. Its gain lands in the body classes rather than the tail, unlike logit adjustment’s.

Does it matter whether the encoder adapts? Only through CUDA. Post-hoc logit adjustment, a fixed prior subtraction at inference, matches every end-to-end method except CUDA, which exceeds it by 0.65 points in point estimate without a paired interval for the difference. Without CUDA, the classifier-only family recovers as much as the end-to-end one. Its share under LoRA, about 0.4, exceeds the quarter it recovered under the frozen readout (Queisler, 2026a), but the two numbers rest on different damages and classifiers. That the one method acting directly on the class prior is also the only one that restores tail recall fits the prior channel’s large share of BRACS’s damage in that study.

Does LoRA itself help? No. LoRA shrinks the measured damage from 7.65 to 4.95 points at $\rho = 100$ only because its balanced model is about three points worse than the frozen readout; its imbalanced models are as accurate as the frozen ones. The best mitigated model at $\rho = 100$, CUDA at 34.17 percent, stays 5.4 points below the frozen balanced readout. At BRACS’s native class shares, LoRA’s damage cannot be distinguished from zero and no rebalancing method differs from CE, so in this pipeline rebalancing matters only at severe imbalance; CUDA’s gain there is better read as a general improvement of the LoRA model.

Limitations. The study covers one dataset, one encoder, and one adaptation scheme. End-to-end training length was tuned on CE at $\rho = 100$ only, and a method or allocation that converges more slowly, such as CUDA under strong augmentation, may be disadvantaged at 600 steps. CUDA was not fitted on the balanced allocation, so its recovery mixes imbalance correction with a general augmentation benefit. Ratio 100 is near the ceiling of this design, so recovery at more extreme imbalance is untested. The full-recovery criterion compares an unpaired interval with a point estimate and is weak whenever the damage is smaller than that interval, as at $\rho = 50$; the paired share is the informative reading. Family means, rank-group recalls, and selection frequencies carry no intervals. The comparison with the frozen readout is descriptive, because the two pipelines differ in classifier and training and share only the test patients.

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